

limits and absolute value

$$\lim_{x \rightarrow -3} \frac{|x+3|}{x+3}$$

$$|x+3| \begin{cases} (x+3) & x \geq -3 \\ -(x+3) & x < -3 \end{cases}$$

$$\frac{(x+3)}{x+3} = 1$$

$$\frac{-(x+3)}{x+3} = -1$$

We have used both of the and got a 1 and -1

the definition of an absolute value is this

$$|x| \begin{cases} x & x \geq 0 \\ -(x) & x < 0 \end{cases}$$

the mean that the absolute value of "x" when x is greater than zero is just (x)

example: $|2| = 2$
 $(2) = 2$

these two are completely the same absolute value of 2 and 2 are two same thing unless you add a negative

the second one means that the absolute value of a negative will give us a positive $| -x | = x$ that is just like saying $-(-x) = x$ same thing right just add a negative which will give us a positive

example: $| -2 | = 2$
 $-(-2) = 2$

Same thing just one is in parentheses and the other is in absolute value

$$\lim_{x \rightarrow 2} \frac{|x-2|}{4-x^2} \quad |x-2| \begin{cases} (x-2) & x \geq 2 \\ -(x-2) & x < 2 \end{cases}$$

$$\lim_{x \rightarrow 2^+} \frac{(x-2)}{(x^2-4)} = \frac{(x-2)}{(x-2)(x+2)} = \frac{1}{x+2} = \frac{1}{4}$$

the positive on the two means that "x" will be approaching from the right or positive which tells us that "x" will be greater than two so we use (x-2)

$$\lim_{x \rightarrow 2^-} \frac{-(x-2)}{(x^2-4)} = \frac{-(x-2)}{(x-2)(x+2)} = \frac{1}{x+2} = \frac{1}{4}$$

the negative on the two tells us that "x" will be approaching from the left side so "x" will be less than two

$$\lim_{x \rightarrow 2} \frac{|x-2|}{4-x^2} \text{ does not exist because we get two different answers}$$

Examples:

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{4 - x}$$

$$\frac{\sqrt{x} - 2}{4 - x} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2}$$

$$\begin{aligned} & (\sqrt{x} - 2)(\sqrt{x} + 2) - \\ & (\sqrt{x})^2 + 2\sqrt{x} - 2\sqrt{x} - 4 \\ & (\sqrt{x})^2 - 4 = x - 4 \end{aligned}$$

$$\frac{(x-4)}{-(x-4)(\sqrt{x}+2)} = \frac{-1}{\sqrt{4}+2} = \boxed{-\frac{1}{4}}$$

$$\lim_{x \rightarrow 0} \frac{7x}{\sqrt{x+9}-3} \cdot \frac{\sqrt{x+9}+3}{\sqrt{x+9}+3}$$

$$\begin{aligned} & (\sqrt{x+9}-3)(\sqrt{x+9}+3) \\ & (\sqrt{x+9})^2 + 3\sqrt{x+9} - 3\sqrt{x+9} - 9 \\ & (\sqrt{x+9})^2 - 9 = x+9-9 = x \end{aligned}$$

$$\frac{7x(\sqrt{x+9}+3)}{x}$$

$$\frac{7(\sqrt{0+9}+3)}{7(3+3)}$$

$$7(6) = \boxed{42}$$

$$\lim_{x \rightarrow 3} \frac{\frac{20}{3x+1} - 2}{x-3}$$

$$= \left[\frac{20}{3x+1} - 2 \right] \cdot \frac{1}{x-3}$$

$$\left(\frac{20}{3x+1} \right) - \left(\frac{2}{1} \right) (3x+1)$$

$$\frac{18-6x}{3x+1} \cdot \frac{1}{x-3}$$

$$\frac{20}{3x+1} - \frac{2(3x+1)}{3x+1}$$

$$\frac{-6(x-x)}{3x+1} \cdot \frac{1}{x-3} = \frac{-6}{3x+1}$$

$$\frac{20-6x-2}{3x+1}$$

$$\frac{-6}{3(3)+1} = \frac{-6}{10} = \frac{-3}{5}$$

$$\boxed{-3/5}$$

$$\lim_{x \rightarrow 0} \frac{\frac{2}{x+5} - \frac{2}{5}}{x} = \left[\frac{2}{x+5} - \frac{2}{5} \right] = \frac{1}{x}$$

$$(5) \left(\frac{2}{x+5} \right) - \left(\frac{2}{5} \right) (x+5) \rightarrow \frac{5(2)}{5(x+5)} - \frac{2(x+5)}{5(x+5)}$$

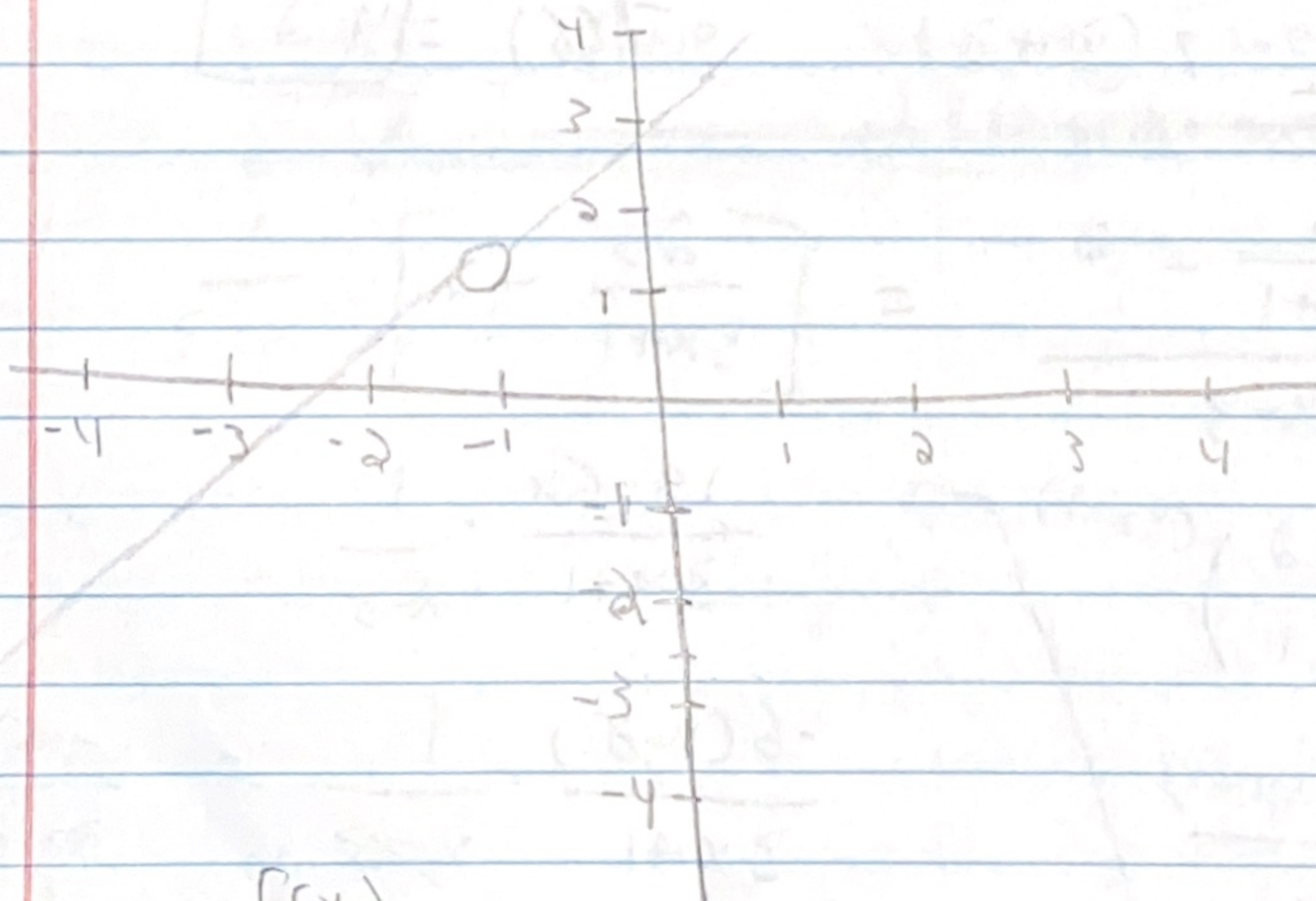
$$\frac{10}{5x+25} - \frac{2x-10}{5x+25} \rightarrow \frac{-2x}{5x+25} = \frac{1}{x}$$

$$\frac{-2}{5(0)+25} = \boxed{\frac{-2}{25}}$$

limit holes

a hole happen when $\lim_{x \rightarrow a} f(x) \neq f(a)$

here is an example:



$$\lim_{x \rightarrow -1} f(x) = 1$$

$$x \rightarrow -1$$

this is a hole $(-1, 1)$

$$f(-1) = \text{DNE}$$